

Pearson Edexcel Award

Tuesday 23 June 2026

Afternoon (Time: 3 hours)

Paper
reference

9811/01

Advanced Extension Award Mathematics

Insert for questions 5, 6 and 7

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5.

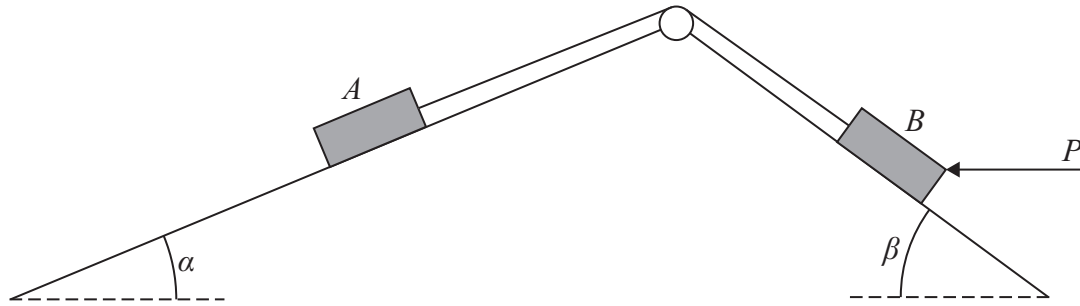


Figure 3

Figure 3 shows a block A with mass $26m$ and a block B with mass $75m$

Block A lies on a rough plane inclined at an angle α to the horizontal.

Block B lies on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth fixed light pulley.

The pulley lies on the line where the two inclined planes meet, with A , B and the pulley all lying in the same vertical plane.

The part of the string between A and the pulley lies along a line of greatest slope of the plane of angle α

The part of the string between B and the pulley lies along a line of greatest slope of the plane of angle β

The two blocks, A and B , are modelled as particles.

- $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{3}{4}$
- the coefficient of friction between block A and its plane is 0.5
- the coefficient of friction between block B and its plane is 0.2

A horizontal force of magnitude P acts on block B . The line of action of P lies in the same vertical plane as A , B and the pulley. The tension in the string is T .

When $k_1 mg \leq P \leq k_2 mg$, the system is at rest in equilibrium with the string taut.

- (a) Determine the exact values of k_1 and k_2 (9)

Assuming that the system remains in equilibrium,

- (b) determine, in simplest form in terms of T , the reaction force acting on the pulley. (3)

Question 5 continued

Without further calculation,

(c) explain the effect of decreasing P below k_1mg on

- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)

Without further calculation,

(d) explain the effect of increasing P above k_2mg on

- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)
(+S2)

(Total for Question 5 is 20 marks)

6. Given that

$$y = \arctan x$$

(a) prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad (2)$$

Given that f is a function such that the integrals exist,

(b) (i) use the substitution $u = a + b - x$ to show that

$$\int_a^b f(x) \, dx = \int_a^b f(a + b - x) \, dx$$

where a and b are constants

(ii) determine the value of the constant λ such that

$$\int_a^b f(x) \, dx = \lambda \int_a^b (f(x) + f(a + b - x)) \, dx \quad (3)$$

Given that $I = \int_{-\pi}^{\pi} \frac{x(1 + \sin x)}{1 + \cos^2 x} \, dx$

(c) use detailed reasoning to show that

$$I = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx \quad (2)$$

Given that

$$g(x) = \frac{x \sin x}{1 + \cos^2 x}$$

(d) show that

$$g(x) = g(-x) \quad (1)$$

(e) Using the results from parts (a), (b), (c) and (d), determine the value of I

(8)
(+S2)

(Total for Question 6 is 18 marks)



7. $y = x^x \quad x > 0$

By taking logarithms of both sides, or otherwise,

- (a) determine $\frac{dy}{dx}$, giving the answer in simplest form in terms of x only. (3)

Given that, for $x > 0$

$$f(x) = (x^x)^x$$

$$g(x) = x^{(x^x)}$$

- (b) determine the exact coordinates of the turning point of $y = f(x)$ (5)
- (c) Show that $g'(x)$ can be written in the form

$$g'(x) = P(x)x^{Q(x)}$$

where $P(x) = x(\ln x)^2 + x \ln x + 1$ and $Q(x)$ is a function of x to be found. (4)

- (d) Solve $P(x) = 1$ (2)

Given that $y = P(x)$ has a stationary point at $x = \alpha$ and a stationary point at $x = \beta$, where $\alpha < \beta$,

- (e) determine the value of α and the value of β (3)

- (f) Hence show that $P(x)$ has a minimum at $x = \beta$ (2)

Using the answers to parts (c), (d), (e) and (f),

- (g) show that $g(x)$ is an increasing function for $x > 0$ (4)
- (+S2)

(Total for Question 7 is 25 marks)



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